

NXT Studio
IoT Data Analytics Internship
Week 2 Assignment

Submitted By

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Course: B.Tech – CSE (INTERNET OF THINGS)

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Submitted To

NXT Studio

Internship Duration

12 Weeks

Assignment

Week 2 - Python for Data Analytics

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Program 1: Variables and Data Types

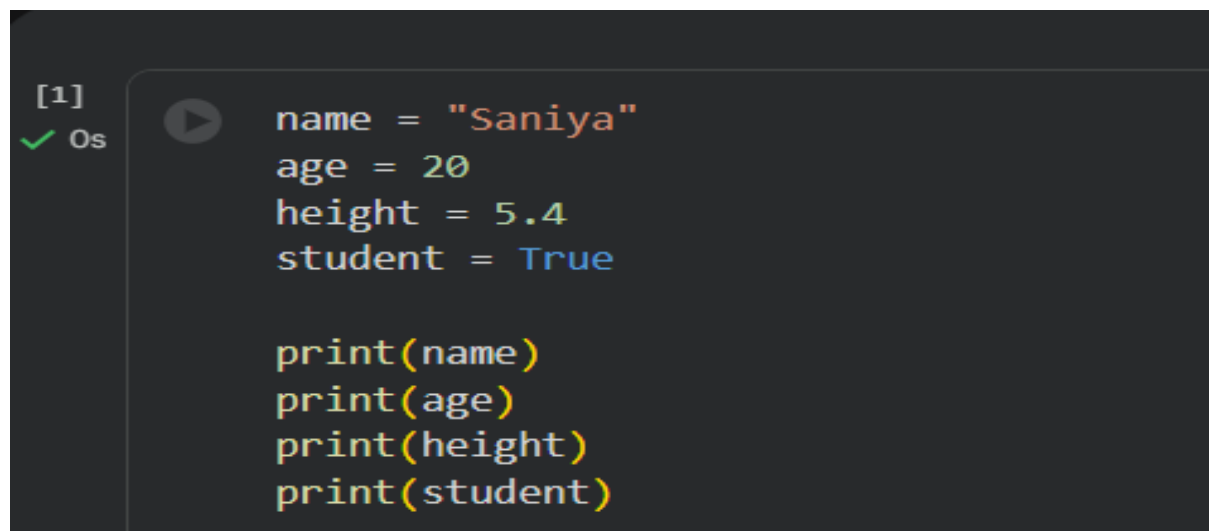
Aim:

To understand variables and different data types in Python.

Theory:

Variables are used to store data in Python. Common data types include int, float, string, and boolean.

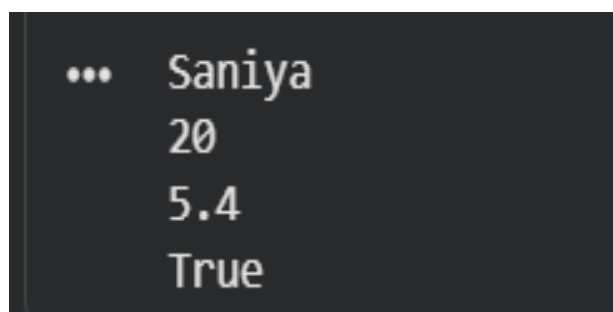
Program:

A screenshot of a code editor with a dark background. On the left, there is a green checkmark and the text "[1] Os". The main area contains the following Python code:

```
name = "Saniya"
age = 20
height = 5.4
student = True

print(name)
print(age)
print(height)
print(student)
```

Output:

A screenshot of the output from the program. It shows four lines of text, each preceded by three dots (the prompt character):

```
... Saniya
... 20
... 5.4
... True
```

Conclusion: The program successfully demonstrates the use of variables and different data types in Python.

Program 2: Loops

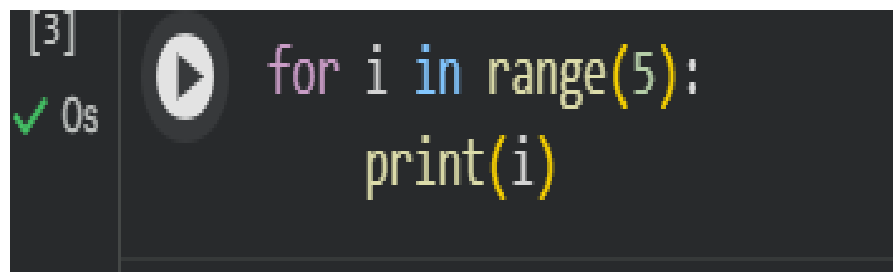
Aim:

To understand the use of loops in python

Theory:

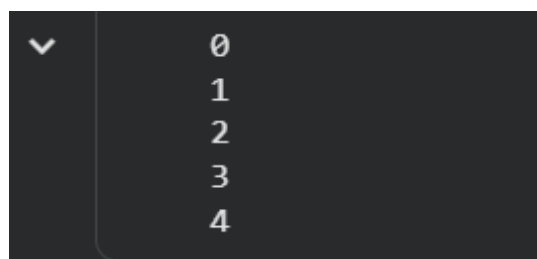
A loop is used to execute a block of code repeatedly. The for loop iterates over a sequence of values.

Program:

A screenshot of a code editor with a dark background. On the left, there is a vertical sidebar with a green checkmark and the text "[3]" and "0s". In the center, there is a play button icon. To the right of the play button, the code "for i in range(5):" is written in a light blue font, and "print(i)" is written in a yellow font below it.

```
[3] 0s for i in range(5):  
      print(i)
```

Output:

A screenshot of the output of the program. It shows a vertical list of numbers from 0 to 4, each on a new line. To the left of the list is a small white downward-pointing arrow.

```
0  
1  
2  
3  
4
```

Conclusion: The program demonstrates the working of a for loop by printing numbers from 0 to 4.

Program 3: Functions

Aim:

To understand the use of functions in Python.

Theory:

A function is a reusable block of code that performs a specific task. Functions help make programs more organized and reduce code repetition.

Program:



```
[4] ✓ 0s def greet(name):  
    print("Hello", name)  
  
    greet("Saniya")
```

Output:



```
▼ Hello Saniya
```

Conclusion: The program demonstrates how to create and call a function in Python.

Program 4: File Handling

Aim:

To understand file handling in Python.

Theory:

File handling allows a program to create, read, write, and modify files. Python provides built-in functions such as `open()`, `write()`, and `read()` for file operations.

Program:

```
[6] ✓ 0s file = open("sample.txt", "r")  
print(file.read())  
file.close()
```

Output:

```
✓ Hello World
```

Conclusion: The program demonstrates how to write data to a file and read it using Python.

Program 5: Exception Handling

Aim:

To understand exception handling in Python.

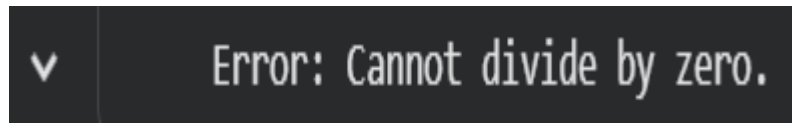
Theory:

Exception handling is used to handle runtime errors without stopping the execution of the program. Python uses try and except blocks to catch and handle exceptions.

Program:

```
[7] ✓ 0s try:  
    number = 10 / 0  
except ZeroDivisionError:  
    print("Error: Cannot divide by zero.")
```

Output:

A dark-themed terminal window with a green checkmark icon on the left. The text "Error: Cannot divide by zero." is displayed in a light green monospace font.

Conclusion: The program successfully demonstrates exception handling by catching a division-by-zero error.

Program 6: Introduction to NumPy

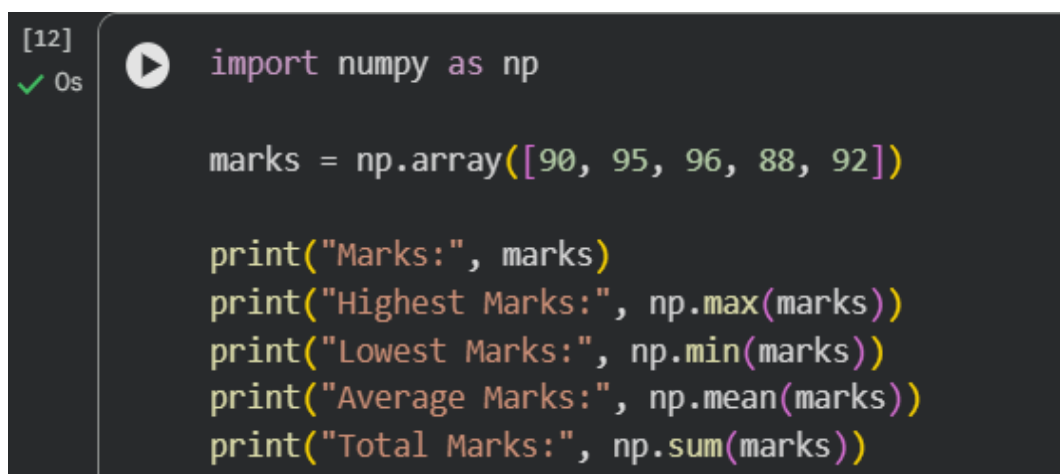
Aim:

To understand the basics of the NumPy library and perform simple array operations.

Theory:

NumPy is a Python library used for numerical computing. It provides support for arrays and mathematical operations, making data processing faster and more efficient.

Program:

A dark-themed code editor window showing Python code. On the left, it says "[12]" and "0s" with a green checkmark. A play button icon is next to the first line of code. The code defines a NumPy array 'marks' and prints its maximum, minimum, mean, and sum.

```
[12]
✓ 0s
import numpy as np

marks = np.array([90, 95, 96, 88, 92])

print("Marks:", marks)
print("Highest Marks:", np.max(marks))
print("Lowest Marks:", np.min(marks))
print("Average Marks:", np.mean(marks))
print("Total Marks:", np.sum(marks))
```

Output:

```
▼ ... Marks: [90 95 96 88 92]
    Highest Marks: 96
    Lowest Marks: 88
    Average Marks: 92.2
    Total Marks: 461
```

Conclusion: The program demonstrates how to create a NumPy array and calculate its average using the mean() function.

Conclusion:

During Week 2: Python for Data Analytics, I learned the fundamentals of Python programming, including variables and data types, loops, functions, file handling, exception handling, and the basics of the NumPy library. I successfully implemented and executed each program and understood their practical applications in data analytics.

References

Python Official Documentation –
<https://docs.python.org/3/>

NumPy Official Documentation –
<https://numpy.org/doc/>